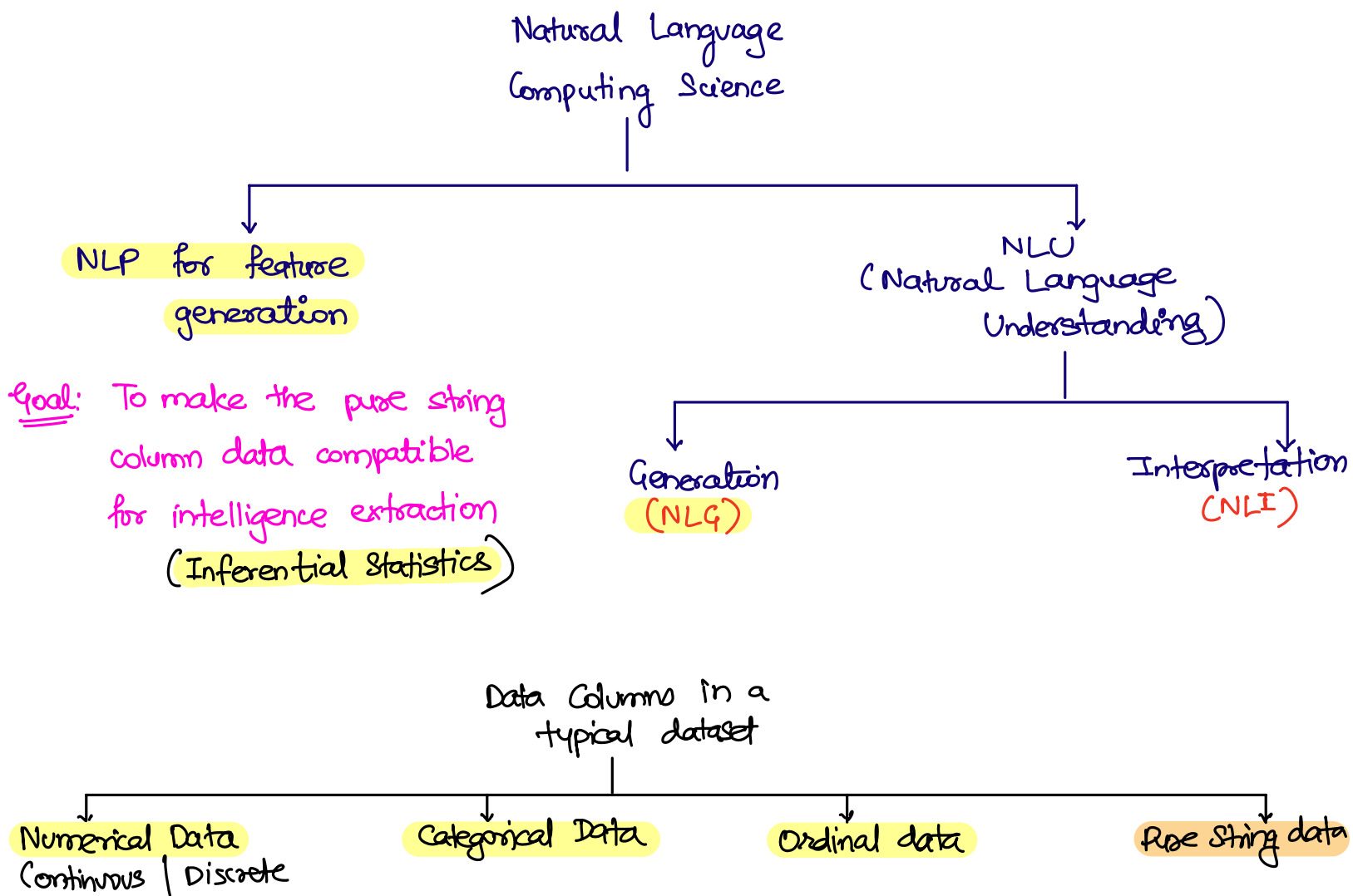


Natural Language Processing

Trainer: Prashant N.

NLP is like teaching computers to READ, TALK, THINK & WRITE in human language.

NLP is the branch of AI that helps
UNDERSTAND,
INTERPRET,
MANIPULATE
the human language.



Text Preprocessing Stage in NLP for feature generation

It has 8 stages. The stages can be skipped if not required for the problem statement.

- Ⓐ Data Acquisition
- Ⓑ Data Cleaning Stage
- Ⓒ Data Normalization Stage
- Ⓓ Data Tokenization Stage
- Ⓔ Stopwords Removal
- Ⓕ POS tagging
- Ⓖ NER (Named Entity Recognition)
- Ⓗ Stemming and Lemmatization

Goal:

make string feature
column compatible
for Inferential Stats.

Ⓐ Data acquisition stage. (Dealing with text data)

This stage is all about collecting/getting RAW DATA and storing the same in a persistent storage.

- file
- Database
- online (XML, JSON, HTML)
- scraping

⑥ Cleaning Stage:

This stage is all about extracting required/relevant data from raw data.

eg: data1.html

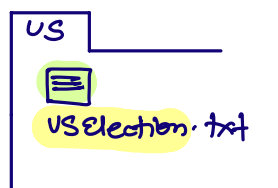
```
<html>
<head>
  <title> US Election </title>
</head>
<body>
  <p> [text] </p>
</body>
</html>
```

Option 1

Define structure and separate data from metadata.

title	article
US Election	[text]

Option 2



Topic Modelling

⑦ Normalization: Bringing the feature column data in one scale

[Text Normalization]

The goal of text normalization is to **UNIFY** the text data.

↑
converting text data into one single case.

eg: Making text follow one case [Uppercase | Lowercase]
Removal of Punctuations.

⑧ Tokenization: Convert text data into a sequence of tokens

Tokenization

Word Tokenization

Set[Paragraph] $\rightarrow f^n \rightarrow$ word List

I am PN

[I, am, PN]

Sentence Tokenization

Set[Paragraph] $\rightarrow f^n \rightarrow$ sentence List

(any NLW activity)

(applied on feature sets
for inferential stats)

⑤ Stopwords Removal:

• they are language specific

• these words have least / NO statistical impact on
AI algo.

eg. english language has diff stopwords
german has diff stopwords

(Outliers
in text
processing)

eg: NLP is Awesome

↓ Normalization (lowercase)

nlp is awesome

↓ stopwords removal

nlp awesome